

Aircraft Optimal Design

MEAer

3rd Assignment2nd semester – 2025/26**Report due date: 7 June**

Handout date: 18 May

Instructions: Install `OpenMDAO` and `OpenAeroStruct` following the instructions included in the appendix of the class notes. This is a prerequisite to complete this assignment. Before the due date, email a brief *report* (**max. 16 pages**) with the results and the *source code* that reproduces the solutions. Add your student numbers as filenames prefix (e.g. `report_123456_234567_345678_456789.pdf` and `code_123456_234567_345678_456789.zip`).

Problem 1 Constrained Optimization using `OpenMDAO` (6 marks)

Consider the minimization of the Himmelblau's function:

$$\min f(\mathbf{x}) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$$

- Determine the analytic partial derivatives of the function $f(\mathbf{x})$ with respect to the variables $\mathbf{x} = (x_1, x_2)$.
- Define the model $f(\mathbf{x})$ in `OpenMDAO` and represent its structure schematically.

Tip: see https://openmdao.org/newdocs/versions/latest/basic_user_guide/single_disciplinary_optimization/first_analysis.html and https://openmdao.org/newdocs/versions/latest/basic_user_guide/command_line/make_n2.html.

- Solve the unconstrained problem numerically, starting from initial guess $\mathbf{x}_0 = (0, 0)$. Instruct `OpenMDAO` to approximate all partial derivatives using the finite-difference method.

Tip: see https://openmdao.org/newdocs/versions/latest/basic_user_guide/single_disciplinary_optimization/first_optimization.html.

- Add the constraint $-x_1 - x_2 \leq 0$ and solve the constrained problem starting from both the initial guess and the optimal solution of c).

Add a profiling capability to the script to count the number of times each function is called and the CPU time it takes.

Tip: see <https://openmdao.org/newdocs/versions/latest/features/debugging/profiling/index.html>.

- e. Run the constrained optimization again but this time using the analytic user-defined derivatives from a).

Compare and comment the performance of each sensitivity analysis approach.

Tip: see https://openmdao.org/newdocs/versions/latest/advanced_user_guide/analytic_derivatives/partial_derivs_explicit.html.

Problem 2 Analysis Models in **OpenAeroStruct** (4 marks)

Answer to the question that applies:

- (Option if IST lowest number student in group is **even**)
OpenAeroStruct uses a low-fidelity aerodynamic model based on the Vortex-Lattice Method (VLM) using a formulation with horse-shoe vortices. With reference to textbooks on Aerodynamics and CFD and based on the python code in directory **openaerostruct/aerodynamics**, explain the theoretical background of this method and describe its numerical implementation in **OpenAeroStruct**.
- (Option if IST lowest number student in group is **odd**)
OpenAeroStruct uses a low-fidelity structural model based on a circular hollow tube using a formulation with 6 degrees of freedom 3-D spatial beams. With reference to textbooks on Structural Mechanics and CSM/FEM and based on the python code in directory **openaerostruct/structures**, explain the theoretical background of this method and describe its numerical implementation in **OpenAeroStruct** (note: consider the 'tube' option only, ignoring the 'wingbox' option also supported).

Include flowchart with relevant routines and I/O arguments.

Problem 3 Aerodynamic Optimization using `OpenAeroStruct`
(10 marks)

Consider the typical aerodynamic optimization problem of an isolated aircraft wing:

$$\begin{aligned} &\text{minimize} && C_D \\ &\text{w.r.t} && \alpha, \gamma(y), c(y) \\ &\text{subject to} && C_L = C_L^* \\ &&& C_l \leq C_{l \max} \\ &&& S_{ref} = S_{ref}^* \end{aligned}$$

where C_D and C_L are the drag and lift coefficients, respectively, α is the angle-of-attack, $\gamma(y)$ is the wing twist spanwise distribution and $c(y)$ is the chord spanwise distribution, and S_{ref} is the planform wing area.

- a. Assuming inviscid drag only, describe the known theoretical optimal solutions and discuss whether these are used in practice.
- b. For the operating conditions and specifications given in Tab.1, find the optimal solution using as design variables:
 - (i) only α ;
 - (ii) α and $\gamma(y)$;
 - (iii) α and $c(y)$;
 - (iv) α , $\gamma(y)$ and $c(y)$.

Summarize the results in tabular and graphical form and discuss them.

Tip: See script `run_aero.py`.

- c. Define a new function related to aerodynamics that can be used as additional constraint and implement it properly in `OpenAeroStruct`, paying attention to its derivatives with respect to its inputs.

Tip: As example, see coding structure of class `LiftDrag` in file `/aerodynamics/lift_drag.py`.

- d. Re-run b. with your new constraint and evaluate its impact on the optimal solution.

Parameter	Value
Cruise speed	$v = 63m/s$
Cruise altitude	$h = 2,000m$

Parameter	Value
Wing type	rectangular
Wing span	$b = 11.0m$
Wing area	$S = 16.2m^2$
Wing section	NACA 2412

Table 1: Cessna 172 data.